

# Solution to the Daily Limit

Based on the problem provided, we are tasked with evaluating the following limit:

$$L = \lim_{x \rightarrow 0} \frac{1}{x^7} \begin{vmatrix} \tan(x) & \sin(x) & x \\ \sinh(x) & \tan^{-1}(x) & x \\ x & \sinh^{-1}(x) & \tanh(x) \end{vmatrix}$$

To find this value, we can use the Maclaurin series expansions of the relevant functions to extract the lowest-order terms of the determinant.

## Rigorous Derivation

### 1. Maclaurin Series Expansions

We expand each trigonometric and hyperbolic function up to  $O(x^5)$  since we are dividing by  $x^7$  and the matrix size is  $3 \times 3$ :

$$\begin{aligned} \tan(x) &= x + \frac{x^3}{3} + \frac{2x^5}{15} + O(x^7) \\ \sin(x) &= x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^7) \\ \sinh(x) &= x + \frac{x^3}{6} + \frac{x^5}{120} + O(x^7) \\ \tan^{-1}(x) &= x - \frac{x^3}{3} + \frac{x^5}{5} + O(x^7) \\ \sinh^{-1}(x) &= x - \frac{x^3}{6} + \frac{3x^5}{40} + O(x^7) \\ \tanh(x) &= x - \frac{x^3}{3} + \frac{2x^5}{15} + O(x^7) \end{aligned}$$

### 2. Factoring the Matrix

Let  $A(x)$  be the  $3 \times 3$  matrix inside the determinant. Notice that every term in  $A(x)$  has a leading factor of  $x$ . We can factor  $x$  out of each of the three columns, which pulls an  $x^3$  out of the determinant:

$$\det(A(x)) = x^3 \det(B(x))$$

where  $B(x) = \frac{1}{x}A(x)$ . The limit then simplifies to finding the  $x^4$  coefficient of  $\det(B(x))$ :

$$L = \lim_{x \rightarrow 0} \frac{x^3 \det(B(x))}{x^7} = \lim_{x \rightarrow 0} \frac{\det(B(x))}{x^4}$$

### 3. Analyzing $B(x)$

Using our expansions, we can write  $B(x)$  as a sum of matrices based on powers of  $x$ :

$$B(x) = J + x^2 B_2 + x^4 B_4 + O(x^6)$$

where  $J$  is the  $3 \times 3$  matrix of all ones, and  $B_2$  is the matrix of  $x^3$  coefficients from our original expansions:

$$B_2 = \begin{pmatrix} 1/3 & -1/6 & 0 \\ 1/6 & -1/3 & 0 \\ 0 & -1/6 & -1/3 \end{pmatrix}$$

#### 4. Evaluating the Determinant

We are looking for the  $x^4$  coefficient of  $\det(J + x^2B_2 + x^4B_4)$ . By the multilinearity of determinants, and because  $J$  is a rank-1 matrix, any determinant formed by taking more than one column from  $J$  is zero.

The  $O(1)$  term is  $\det(J) = 0$ . The  $O(x^2)$  term requires taking two columns from  $J$  and one from  $B_2$ , which is also 0. The  $O(x^4)$  term is obtained by taking exactly one column from  $J$  and two columns from  $B_2$ . This is precisely the sum of all elements of the adjugate matrix of  $B_2$ , which can be written as  $\text{Tr}(\text{adj}(B_2)J)$ .

Let us calculate the cofactors  $C_{ij}$  of  $B_2$ :

$$\begin{aligned} C_{11} &= \frac{1}{9}, & C_{12} &= \frac{1}{18}, & C_{13} &= -\frac{1}{36} \\ C_{21} &= -\frac{1}{18}, & C_{22} &= -\frac{1}{9}, & C_{23} &= \frac{1}{18} \\ C_{31} &= 0, & C_{32} &= 0, & C_{33} &= -\frac{1}{12} \end{aligned}$$

Summing all the cofactors gives the coefficient of  $x^4$ . We break this sum across multiple lines so it fits perfectly within the page margins:

$$\begin{aligned} \sum_{i,j=1}^3 C_{ij} &= \left(\frac{1}{9} - \frac{1}{18} + 0\right) + \left(\frac{1}{18} - \frac{1}{9} + 0\right) + \left(-\frac{1}{36} + \frac{1}{18} - \frac{1}{12}\right) \\ &= 0 + 0 + \left(-\frac{1}{36} + \frac{2}{36} - \frac{3}{36}\right) \\ &= -\frac{2}{36} \\ &= -\frac{1}{18} \end{aligned}$$

#### Conclusion

Since the determinant of  $B(x)$  can be expanded as  $\det(B(x)) = -\frac{1}{18}x^4 + O(x^6)$ , the limit evaluates perfectly to:

$$L = -\frac{1}{18}$$